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Performance analysis and comparison of crested porcupine optimization algorithm with state-of-the-art metaheuristics

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Abstract:

The Crested Porcupine Optimization (CrPO) algorithm is a recently developed nature-inspired metaheuristic based on the distinct multi-phase defense mechanisms observed in crested porcupines. This work evaluates the algorithm's effectiveness using the CEC 2015 benchmark set, which comprises various test functions, including unimodal, multimodal, hybrid, and composition categories. CrPO introduces a population reduction strategy that operates cyclically, alongside four novel defense-modeled procedures, aiming to maintain a strong balance between exploration and exploitation phases. To assess its performance, the algorithm was benchmarked against four contemporary metaheuristics: Differential Evolution (DE), Love Evolution Algorithm (LEA), Hippopotamus Optimization (HipO), Covariance Matrix Adaptation Evolution Strategy (CMA-ES), Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and the Aquila Optimizer (AO). Each test was conducted on both 10-dimensional and 30-dimensional instances, with 51 independent runs per problem to ensure statistical reliability. The empirical results indicate that CrPO outperforms its counterparts on the majority of test cases, securing the lowest (best) mean rank across both dimensional scales. Additionally, the superiority of CrPO was confirmed through statistical analysis using Friedman's mean rank test, reinforcing its potential as a robust solution approach for tackling challenging numerical optimization tasks in scientific and engineering contexts.

Keywords: *CEC 2015, Crested porcupine optimization, Meta-Heuristics, Optimization*

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1. INTRODUCTION

Most metaheuristics focus on solving optimization problems inspired by metaphors that exist in nature [1]. The number of such algorithms is progressively increasing in the literature. Walrus Optimization Algorithm (WO) was developed based on the struggle of walruses to survive as a herd in deep water environments, which are their natural habitat [2]. This is inspired by the four basic behaviors of walruses: Migration, breeding, roosting, and foraging. Whether being male, female or young causes these individuals to show behavioral differences depending on the threat situation. Love Evolution Algorithm (LEA) was inspired by the Stimulus-Value-Role (SVR) theory [3, 4]. According to this theory, there are phases in which people determine their preferences in choosing a couple for marriage. The first one is called the stimulus phase, and it is the phase in which two people do not interact directly with each other, but establish a bond based on the impressions they obtain via visual and auditory elements. At this phase, appearance, behavioral and personal characteristics are taken into consideration. In the phase of value, it is known that if the process of valuing mutual opinions by talking ends negatively for any person, the relationship will not progress to the next stage. In the role part, which is the last stage, two individuals mutually make a judgment about whether they can fulfill the roles that the relationship sets for both partners. In short, it is a method that adapts the optimization of couples to living a happy life based on human character traits such as temperament, personality and hobbies, in a way that can produce solutions within the objective function. Hippopotamus optimization (HiPO), as the name suggests, is a meta-heuristic algorithm inspired by the instinctive behavior of hippos [5]. These behaviors are basically defined as three phases. In the first phase, hippopotamuses live in groups. Each group consists of calves, females, and adult males, one of whom is the leader in the group. In this phase, young hippopotamuses may leave the herd since their curiosity and become the target of predators. The second phase involves the defensive behavior of hippopotamuses that are attacked by a predator. In such a situation, turning around the predator, showing its majestic jaws and making various sounds are the behaviors of hippopotamuses to drive the predators away. In the last phase, hippopotamuses try to escape from predators by running away and entering the nearest lake or pond. Crested porcupine optimization (CrPO) derives from the highly effective defensive abilities of a beaver-like rodent, also known as the crowned porcupine, against predators [6]. The behavior of this species is examined in four different zones depending on the distance from a predator such as a leopard. The first zone involves the porcupine raising its spines and trying to make itself look bigger than it is, as a defensive behavior when the hunter is at its furthest distance when it is first noticed. If this behavior does not scare the predator and it gets closer, the second zone of defense comes into play. In the second zone, the crested porcupine aims to scare its predator by making various sounds. If the predator is not afraid of this technique and continues to approach, the next form of defense is the fetid chemical odor strategy. If this strategy does not work, the last step is to shoot arrows — to launch sharp quills as a defensive projectile mechanism — when the hunter animal is closest to the crested porcupine. This technique usually injures or kills the predator. In addition to the optimization part based on these four defensive behaviors, a method called cyclic population reduction technique is used. It is claimed by the authors that this method accelerating convergence speed and preserves population diversity. This process can be summarized as removing some individuals from the population and then reintroducing them.

Similar to the mentioned algorithms above, some of the studies carried out in recent years on metaheuristic optimization methods inspired by nature are as follows: life cycle of termites [7], some phenomena based on physics [8], working principle of optical microscope [9], foraging and storage strategy of nutcracker birds [10], position and velocity calculation based on Kepler's laws of planetary motion [11], behavior of crayfish [12], ways of the mammal called coati to protect itself from hunting and predators [13], mate-finding behavior of snakes [14], hunting and fighting behavior of the Siberian tiger [15], some rules used in the calculations of vectors [16], and hunting strategies of aquila birds [17].

This study is dedicated to the performance analysis and empirical evaluation of the Crested Porcupine Optimization (CrPO) algorithm [6], which was recently introduced in the literature. We emphasize that

this work does not propose a new algorithm; rather, it assesses the efficacy and scalability of CrPO's unique methodological features in a rigorous comparative setting. CrPO is specifically chosen for this analysis due to its distinct algorithmic contributions, which extend beyond its biological metaphor, notably including a cyclical population reduction strategy and adaptive multi-stage defense transitions. The effectiveness of CrPO and its competitive standing against state-of-the-art metaheuristic methods were evaluated using the comprehensive CEC 2015 benchmark suite [18], a widely recognized and challenging platform for testing numerical optimization algorithms. This rigorous benchmarking aims to establish the true performance profile of CrPO for the wider optimization community.

2. MATERIAL AND METHODS

2.1. Crested Porcupine Optimization Algorithm

Crested porcupines (CPs), inhabit all continents except Antarctica and belong to the category of sizable rodents. They generally live-in deserts, forests, hillsides, and rocky areas. CPs use various defense strategies against predators. These strategies can be increasingly aggressive depending on the threat level. The first defensive strategy (sight) is to make the CP appear larger by erecting its long and sharp spines. The second defense (sound) is that they make various sounds when they feel threatened, they can make high-pitched sounds with special spines on their tail. Thirdly (odor), CPs also secrete a bad smell and try to repel predators by spreading this smell with special hairs on their back. Finally (physical attack), they can attack physically, thus causing serious injuries by stabbing their thick spines into the skin of their enemies.

Crested Porcupine Optimization (CrPO) is a metaheuristic based on defensive behaviors of crested porcupines by Basset et al. in 2024. CrPO mimics the defensive strategies of prey. The algorithm has the steps of initialization, cyclic population reduction, exploration and exploitation. The pseudocode and flowchart of CrPO are given in Algorithm 1.

Algorithm 1. Pseudocode of Crested Porcupine Optimization

Set parameters: N', T_{max}, α, T_f, T, N_{min}
Initialize the solutions' positions X^i randomly, for i = 1 to N'
While (t < T_{max}) do
Evaluate fitness of each candidate solution
Determine current best solution: $\overrightarrow{x_{CP}^t}$
Update γ_t using Eq. (9)
Update population size using Eq. (2)
For i = 1 to N do
Update mass (m), diffusion factor (S), force (F), and $\vec{\delta}$
Generate two random numbers: τ_8 and τ_9
If $\tau_8 < \tau_9$ then
Generate τ_6 and τ_7
If $\tau_6 < \tau_7$ then
Eq. (3) is applied. Else
Else
Use Eq. (5)
Else
Generate τ_{10}
If $\tau_{10} < T_f$ then
Use Eq. (7)
Else
Apply Eq. (11)

End If
End If
If $f\left(\overrightarrow{x_i^{t+1}}\right) > f\left(\overrightarrow{x_i^t}\right)$then
$\overrightarrow{x_i^{t+1}} = \overrightarrow{x_i^t}$
End If
Increment t by 1
End For
End While
Return $\overrightarrow{x_{CP}^t}$

Initialization step: At the outset of the CrPO algorithm, candidate solutions are initialized by selecting values at random from within the specified limits of the search domain. This randomized initialization helps ensure that the population begins with a broad and diverse distribution across the solution space. Each member of this population encodes a possible answer to the optimization problem. By relying on randomness during this setup, the algorithm promotes a broad and varied coverage of the solution landscape, laying the groundwork for thorough exploration at the outset of the search process.

$$\overrightarrow{X_i} = \vec{L} + \vec{r} \cdot \left(\vec{U} - \vec{L}\right), i = 1,2, \dots, N' \quad (1)$$

Equation (1) includes several key parameters: N indicates the total number of individuals within the population, while L and U define the problem's lower and upper boundaries. The term r refers to a vector composed of random values, each independently drawn from a uniform distribution over the interval [0, 1].

Cyclic population reduction technique: To improve convergence efficiency while preserving diversity among candidate solutions, the CrPO algorithm employs a cyclic strategy that periodically reduces the population size. By modifying the number of individuals during different phases of the optimization, this mechanism supports a dynamic shift between broad exploration and focused exploitation as the search progresses.

$$N = N_{min} + (N' - N_{min}) \cdot \left(1 - \left(\frac{t \,\%\, \frac{T_{max}}{T}}{\frac{T_{max}}{T}}\right)\right) \quad (2)$$

According to Equation (2), N denotes the current number of individuals in the population, N_{min} represents the minimum allowed population size, T is the cyclic control parameter, t is the current number of function evaluations, and T_{max} indicates the maximum number of function evaluations.

Exploration phase: Within the CrPO framework, two biologically grounded strategies—sight and sound—regulate the algorithm's exploratory behavior. These approaches are inspired by the initial defensive responses of crested porcupines when detecting distant threats. The sight mechanism reflects visually triggered reactions, whereas the sound mechanism models auditory warning cues. Both are designed to steer the search process toward diverse areas of the solution space and help mitigate the risk of early stagnation. By incorporating these behaviors, CrPO strengthens its ability to conduct a broad global search, especially in complex multimodal environments.

From a mathematical standpoint, these behaviors are formulated in Equations (3) and (6). The sight-driven movement directs individuals to either move closer to or away from the best solution found so far, based on stochastic influence, thus encouraging variability. Meanwhile, the sound-based operation perturbs positions by leveraging the difference between randomly chosen solutions, which aids in

expanding the overall search coverage. Collectively, these strategies form the foundation of CrPO's global exploration phase.

First defensive strategy (Sight): In this phase of the algorithm, normally distributed random values are used to influence the direction in which individuals move. When the generated parameter τ_1 falls outside the interval [-1, 1], the crested porcupines (CPs) are assumed to advance toward the perceived threat—mimicking an assertive or risk-tolerant behavior. Conversely, when τ_1 remains within the range, the movement is directed away from the source, resembling a more defensive or evasive reaction. This stochastic rule allows the algorithm to dynamically shift between intensifying the search near promising regions and dispersing toward unexplored areas, reducing the risk of getting trapped in local optima.

$$\overrightarrow{x_i^{t+1}} = \overrightarrow{x_i^t} + \tau_1 \times \left|2 \times \tau_2 \times \overrightarrow{x_{CP}^t} - \overrightarrow{y_i^t}\right| \quad (3)$$

In Equation (3), X_{CP} represents the best solution found so far. The parameter τ_1 is a random value generated from a normal distribution, while τ_2 is a uniformly distributed random number in the interval [0,1]. The term y_i denotes the midpoint between the current crested porcupine (CP) and a randomly selected individual from the population. It is computed as follows:

$$\overrightarrow{y_i^t} = \frac{\overrightarrow{x_i^t} + \overrightarrow{x_r^t}}{2} \quad (4)$$

where x_i^t is the position of the current individual and x_r^t is the position of a randomly selected individual at iteration *t*.

Second defensive strategy (Sound): In this stage, the algorithm mimics the crested porcupine's (CP's) sound-based defensive behavior. CPs emit a variety of warning sounds to deter predators when visual deterrence fails. In the CrPO algorithm, this mechanism is modeled through directional perturbation between two randomly selected individuals in the population. The following equation governs this behavior:

$$\overrightarrow{x_i^{t+1}} = \left(1 - \vec{U}_1\right) \times \overrightarrow{x_i^t} + \vec{U}_1 \times \left(\vec{y} + \tau_3 \times \left(\overrightarrow{x_{r1}^t} - \overrightarrow{x_{r2}^t}\right)\right) \quad (5)$$

In Equation (5), r1 and r2 are randomly selected integers within the range [0, N], representing indices of individuals in the population. The parameter τ_3 is a uniformly distributed random number in the interval [0, 1]. The vector $\vec{U}_1$ is a binary vector composed of elements taking values in {0, 1}, generated to simulate different possible activation states for perturbation.

Exploitation phase: The exploitation phase of the CrPO algorithm is governed by two biologically inspired defense mechanisms: odor and physical attack, which correspond to the third and fourth defensive strategies, respectively. These mechanisms are activated when the predator (i.e., the optimization problem) is perceived to be in close proximity, prompting the crested porcupine to employ more aggressive tactics. In the context of the algorithm, these strategies enhance local search capability by intensifying the exploitation of promising regions in the search space.

Third defensive strategy (Odor): In this stage, the algorithm simulates the odor-based defense mechanism exhibited by crested porcupines. When visual and auditory warnings fail, the animal releases a foul-smelling chemical to repel predators. In CrPO, this behavior is mathematically modeled by diffusing the solution based on its fitness value and a time-adaptive scaling factor. The corresponding

movement is governed by Equation (6), while the diffusion strength and decay over iterations are defined in Equation (7).

$$\overrightarrow{x_i^{t+1}} = \left(1 - \vec{U}_1\right) \times \overrightarrow{x_i^t} + \vec{U}_1 \times \left(\vec{y} + \tau_3 \times \left(\overrightarrow{x_{r1}^t} - \overrightarrow{x_{r2}^t}\right)\right) \quad (6)$$

In this context, *r3* is a randomly selected integer from the range [1, N], representing an index within the population. The parameter $\vec{\delta}$ is used to determine the direction and magnitude of the search and is calculated using Equation (8). The defense coefficient $\gamma_t$controls the intensity of the odor-based response and is adaptively updated over time using Equation (9). The variable τ is a random number uniformly distributed in the interval [0, 1]. The term $\vec{S}_i^t$ denotes the diffusion factor associated with the i-th individual at iteration *t*, and it is computed using Equation (10).

$$\overrightarrow{x_i^{t+1}} = \left(1 - \vec{U}_1\right) \times \overrightarrow{x_i^t} + \vec{U}_1 \times \left(\overrightarrow{x_{r1}^t} + S_i^t \times \left(\overrightarrow{x_{r2}^t} - \overrightarrow{x_{r3}^t}\right)\right) - \tau_3 \times \vec{\delta} \times \gamma_t \times S_i^t \quad (7)$$

$$\vec{\delta} = \begin{cases} +1, & if\ \vec{rand} \leq 0.5 \\ -1, & Else \end{cases} \quad (8)$$

$$\gamma_t = 2 \times rand \times \left(1 - \frac{t}{t_{max}}\right)^{\frac{t}{t_{max}}} \quad (9)$$

$$S_t^i = \exp\left(\frac{f(x_t^i)}{(\sum_{k=1}^{N} f(x_t^k) + \varepsilon)}\right) \quad (10)$$

The equations utilize the rand function to generate real-valued samples drawn from a uniform distribution spanning the range [0,1], which supports consistent stochastic behavior during the algorithm's execution. The parameter ε is a small positive constant introduced to prevent division by zero. The term $f(x_t^i)$ represents the objective function value of the i-th individual at iteration *t*, and it is used to guide the search direction and intensity during the exploitation process.

Fourth defensive strategy (Physical Attack): This strategy reflects the final and most aggressive defensive behavior of crested porcupines when a predator is in close proximity. In such cases, the animal initiates a physical attack using its sharp quills. In the CrPO algorithm, this behavior is modeled by combining momentum-based movement and a force term influenced by the fitness and mass of each individual. The movement equation includes components representing the applied force and velocity variation, enhancing the local exploitation ability of the algorithm. The overall update mechanism is formulated in Equation (11), which enables precise adjustments in high-quality regions of the search space and contributes to faster convergence.

$$\overrightarrow{x_i^{t+1}} = \overrightarrow{x_{CP}^t} + (\alpha(1 - \tau_4) + \tau_4) \times \left(\delta \times \overrightarrow{x_{CP}^t} - \overrightarrow{x_i^t}\right) - \tau_5 \times \delta \times \gamma_t \times \overrightarrow{F_i^t} \quad (11)$$

In Equation (11), $\vec{x}_{CP}^t$ represents the best solution found so far in the population. The parameter α is a convergence control factor that regulates the step size during exploitation. The variable τ_4 is a random number generated uniformly in the interval [0, 1]. The term $\vec{F}_i^t$ denotes the average force acting on the i-th individual and is calculated using Equation (12), contributing to the intensity of the physical attack mechanism based on solution quality.

$$\overrightarrow{F_i^t} = \vec{\tau}_6 \times \frac{\left(m_i \times \left(\overrightarrow{v_i^{t+1}} - \overrightarrow{v_i^t}\right)\right)}{\Delta t} \quad (12)$$

$$m_i = e^{\frac{f(\vec{x_i^t})}{\left(\sum_{k=1}^{N} f(\vec{x_k^t}) + \epsilon\right)}} \quad (13)$$

$$\overrightarrow{v_i^t} = \overrightarrow{x_i^t}, \quad (14)$$

$$\overrightarrow{v_i^{t+1}} = \overrightarrow{x_r^t},$$

In this context, m_i represents the mass of the i-th individual, which is typically derived from its fitness value. The function *F(.)* denotes the objective (fitness) function being minimized or maximized. The term $\vec{v}_i^{t+1}$ refers to the updated velocity of the i-th individual at iteration *t+1*, while $\vec{v}_i^t$ denotes its velocity at the current iteration *t*. These velocity components are used to simulate the momentum aspect of the physical attack behavior, enabling dynamic movement and precise exploitation in the local region of the search space. The parameter $\vec{\tau}_6$ represents a random value drawn from a uniform distribution over the interval [0,1].

3. RESULTS AND DISCUSSION

The effectiveness of the Crested Porcupine Optimizer (CrPO) was assessed using the CEC 2015 benchmark suite, which includes 15 diverse test functions designed to represent a range of optimization challenges—such as unimodal, simple multimodal, hybrid, and composition problems [18]. The names and properties of these functions are listed in Table 1. The experimental results of CrPO were compared against seven well-known algorithms: Aquila Optimizer (AO) [17], Differential Evolution (DE) [19] Hippopotamus Optimization Algorithm (HipO) [5], Love Evolution Algorithm (LEA) [3], Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [20], Particle Swarm Optimization (PSO) [21] and, Genetic Algorithm [22]. All algorithms were evaluated in both 10-dimensional and 30-dimensional problem settings, as specified by the benchmark standards. Each algorithm was independently executed 51 times for each test function. The computational budget was set to *Dx10000* function evaluations, where *D* denotes the problem dimension. For all experiments, the lower and upper bounds of the decision variables were set to [-100, 100]. The statistical metrics considered in this study comprise the best, worst, mean, and standard deviation of the results, as well as the p-value for significance testing, the ranking of algorithms according to their mean performance, and the computational time required for execution. In addition, Friedman's mean rank test was conducted to statistically validate the performance differences among the compared algorithms. All experiments were conducted on a machine equipped with an AMD Ryzen 5600 processor and 32 GB of RAM. To ensure fairness, the population size for all algorithms was fixed at 30. Other control parameters were taken from the original publications of each algorithm.

Table 1. CEC 2015 learning-based benchmark suite (Ro: Rotated, Sh: Shifted, Hy: Hybrid, Co: Composition, Fu: Function)

No.	Functions (Search Range: [-100, 100]^D)	Fi∗=Fi(x∗)
Unimodal Fu.		
1	Ro. High Conditioned Elliptic Fu.	100
2	Ro. Cigar Fu.	200
Simple Multimodal Fu.		
3	Sh. and Ro. Ackley's Fu.	300
4	Sh. and Ro. Rastrigin's Fu.	400
5	Sh. and Ro. Schwefel's Fu.	500
Hy. Fu.		
6	Hy. Fu.1 (N=3)	600
7	Hy. Fu.2 (N=4)	700
8	Hy. Fu.3 (N=5)	800
Co. Fu.		
9	Co. Fu.1 (N=3)	900

10	Co. Fu.2 (N=3)	1000
11	Co. Fu.3 (N=5)	1100
12	Co. Fu.4 (N=5)	1200
13	Co. Fu.5 (N=5)	1300
14	Co. Fu.6 (N=7)	1400
15	Co. Fu.7 (N=10)	1500

The experimental results for the 10-dimensional test cases were first obtained for CrPO and the seven comparison algorithms. Tables 2 and 3 present the best, worst, average, and standard deviation results obtained from the objective function evaluations, offering a comprehensive view of the algorithm's performance across test cases. CrPO consistently outperformed all competitor algorithms on all Unimodal, Multimodal, and Hybrid test functions. In addition, it ranked first on five of the Composition-type functions. According to Friedman's mean rank test, CrPO achieved the best overall performance with a mean rank of 1.07. It was followed by Differential Evolution (DE) with a mean rank of 2.53, and the Love Evolution Algorithm (LEA) with 2.87. The Hippopotamus Optimizer (HipO) and Particle Swarm Optimization (PSO) had mean ranks of 4.40 and 4.40, respectively, placing them in the fourth and fifth positions. Furthermore, convergence curves for representative functions from each function category (Unimodal, Multimodal, Hybrid, and Composition) are illustrated in Figure 1, highlighting the superior convergence behavior of CrPO. Furthermore, the p-value obtained from the Friedman test was found to be below the 0.05 significance level, indicating that the superior performance of the CrPO algorithm is statistically significant.

The performance results for the 30-dimensional test problems are presented in Table 4 and Table 5, where the best, worst, mean, and standard deviation values of each algorithm are listed. Convergence behaviors are illustrated in Figure 2. CrPO demonstrated a highly competitive performance by outperforming the compared algorithms on 13 out of the 15 benchmark functions. This result confirms the algorithm's robustness and scalability when handling high-dimensional optimization problems. According to Friedman's mean rank test, the overall average rankings of the algorithms were as follows: CrPO ranked first with a mean rank of 1.13, followed by DE with 2.67, LEA with 2.87, HiPO with 4.40, and GA with 4.47. These results reinforce the consistency and superior global search capabilities of CrPO as the problem dimensionality increases. Furthermore, the p-value obtained from the Friedman test was found to be below the 0.05 significance level, indicating that the superior performance of the CrPO algorithm is statistically significant.

Table 2. CEC 2015 – 10-dim-1

	Metric	AO	DE	PSO	HIPO	LEA	CMAES	GA	CrPO
Fnc1	worst	2.568E+07	1.061E+06	1.382E+07	9.023E+06	5.009E+05	3.970E+07	7.432E+07	3.909E+01
	mean	1.516E+07	1.148E+05	2.225E+06	4.138E+06	1.639E+05	1.572E+07	2.336E+07	3.796E+00
	best	6.974E+06	9.646E+00	1.200E+05	8.702E+04	1.922E+04	5.187E+06	6.261E+05	6.425E-03
	std	5.477E+06	2.958E+05	3.646E+06	2.336E+06	1.218E+05	1.032E+07	2.373E+07	1.011E+01
	time	2.637E-01	4.137E-01	3.508E+02	1.579E+00	7.530E-01	1.734E+00	7.639E-01	2.745E-01
	p-val	3.125E-08	4.159E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	6.000E+00	2.000E+00	4.000E+00	5.000E+00	3.000E+00	7.000E+00	8.000E+00	1.000E+00
Fnc2	worst	2.504E+09	2.699E+06	2.745E+09	3.391E+04	3.020E+04	1.064E+10	5.750E+08	1.076E-15
	mean	1.329E+09	1.351E+05	5.060E+08	7.260E+03	8.642E+03	4.911E+09	1.079E+08	1.215E-16
	best	1.999E+08	2.501E+00	1.774E+07	3.185E+02	3.144E+01	8.622E+05	1.958E+06	1.122E-21
	std	5.753E+08	5.733E+05	7.885E+08	1.033E+04	8.663E+03	2.398E+09	1.407E+08	2.737E-16
	time	1.601E-01	3.129E-01	4.564E+02	1.077E+00	4.783E-01	1.223E+00	5.413E-01	1.440E-01
	p-val	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	7.000E+00	4.000E+00	6.000E+00	2.000E+00	3.000E+00	8.000E+00	5.000E+00	1.000E+00
Fnc3	worst	2.060E+01	2.035E+01	2.023E+01	2.006E+01	2.008E+01	2.069E+01	2.018E+01	2.000E+01
	mean	2.042E+01	2.024E+01	2.008E+01	2.002E+01	2.001E+01	2.055E+01	2.004E+01	1.905E+01
	best	2.020E+01	2.009E+01	2.002E+01	2.000E+01	2.000E+01	2.034E+01	2.000E+01	0.000E+00
	std	8.761E-02	6.031E-02	5.643E-02	1.515E-02	2.114E-02	1.011E-01	3.814E-02	4.259E+00
	time	2.071E-01	3.242E-01	4.502E+02	1.077E+00	4.905E-01	1.223E+00	5.491E-01	1.513E-01
	p-val	3.125E-08	3.125E-08	3.125E-08	3.125E-08	5.176E-04	3.109E-08	3.125E-08	ref

	rank	7.000E+00	6.000E+00	5.000E+00	3.000E+00	2.000E+00	8.000E+00	4.000E+00	1.000E+00
Fnc4	worst	7.043E+01	1.244E+01	3.087E+01	4.179E+01	3.413E+01	8.197E+01	4.517E+01	9.950E+00
	mean	5.206E+01	6.546E+00	2.236E+01	3.078E+01	2.215E+01	3.223E+01	2.640E+01	5.022E+00
	best	4.021E+01	2.985E+00	1.760E+01	8.955E+00	1.094E+01	1.974E+01	1.291E+01	0.000E+00
	std	8.227E+00	2.385E+00	3.169E+00	8.255E+00	6.645E+00	1.584E+01	8.483E+00	2.466E+00
	time	1.900E-01	3.098E-01	4.777E+02	1.033E+00	4.891E-01	1.211E+00	5.521E-01	1.594E-01
	p-val	3.050E-08	5.374E-02	3.050E-08	3.521E-08	3.050E-08	3.050E-08	3.050E-08	ref
	rank	8.000E+00	2.000E+00	4.000E+00	6.000E+00	3.000E+00	7.000E+00	5.000E+00	1.000E+00
Fnc5	worst	1.887E+03	1.169E+03	8.781E+02	1.213E+03	1.391E+03	1.952E+03	1.139E+03	3.638E+02
	mean	1.471E+03	3.343E+02	4.476E+02	8.022E+02	6.427E+02	1.684E+03	6.707E+02	1.297E+02
	best	8.585E+02	3.042E+01	1.593E+02	3.826E+02	2.072E+02	1.123E+03	2.079E+02	3.602E+00
	std	3.189E+02	2.756E+02	1.714E+02	2.150E+02	2.874E+02	1.983E+02	2.687E+02	1.180E+02
	time	2.254E-01	3.124E-01	4.744E+02	1.066E+00	5.121E-01	1.222E+00	5.574E-01	2.046E-01
	p-val	3.125E-08	2.148E-03	9.325E-07	3.125E-08	9.659E-08	3.125E-08	1.460E-07	ref
	rank	7.000E+00	2.000E+00	3.000E+00	6.000E+00	4.000E+00	8.000E+00	5.000E+00	1.000E+00
Fnc6	worst	4.599E+05	4.706E+02	1.067E+06	9.827E+04	8.987E+03	4.483E+06	4.237E+06	1.316E+01
	mean	1.153E+05	1.078E+02	2.072E+05	1.534E+04	3.901E+03	1.110E+06	5.567E+05	4.598E+00
	best	9.202E+03	1.238E+00	9.798E+02	1.104E+03	5.511E+02	1.537E+04	1.156E+04	2.845E-12
	std	1.192E+05	1.277E+02	4.167E+05	2.634E+04	2.785E+03	1.149E+06	9.101E+05	5.102E+00
	time	1.995E-01	3.244E-01	4.763E+02	1.101E+00	4.813E-01	1.229E+00	5.645E-01	1.533E-01
	p-val	3.125E-08	1.204E-06	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	5.000E+00	2.000E+00	6.000E+00	4.000E+00	3.000E+00	8.000E+00	7.000E+00	1.000E+00
Fnc7	worst	1.805E+01	1.315E+00	1.374E+01	6.464E+00	4.150E+00	2.093E+01	5.781E+00	5.025E-01
	mean	6.845E+00	4.906E-01	4.629E+00	4.277E+00	2.531E+00	1.560E+01	3.831E+00	1.538E-01
	best	3.751E+00	7.396E-03	2.316E+00	2.145E+00	1.245E+00	6.665E+00	1.804E+00	1.943E-02
	std	3.691E+00	4.352E-01	2.454E+00	1.180E+00	7.851E-01	3.632E+00	1.147E+00	1.233E-01
	time	3.365E-01	4.199E-01	5.714E+02	1.241E+00	6.490E-01	1.312E+00	6.861E-01	2.680E-01
	p-val	3.121E-08	6.589E-03	3.121E-08	3.121E-08	3.121E-08	3.121E-08	3.121E-08	ref
	rank	7.000E+00	2.000E+00	6.000E+00	5.000E+00	3.000E+00	8.000E+00	4.000E+00	1.000E+00
Fnc8	worst	3.604E+04	2.019E+02	2.556E+04	2.992E+03	5.372E+03	3.679E+05	2.996E+06	1.682E+01
	mean	9.518E+03	3.596E+01	3.367E+03	6.969E+02	1.909E+03	1.498E+05	5.526E+05	2.822E+00
	best	9.339E+02	3.380E-01	2.109E+02	7.583E+01	3.518E+02	3.550E+03	6.298E+02	3.618E-04
	std	7.724E+03	5.914E+01	7.325E+03	5.671E+02	1.394E+03	1.444E+05	9.940E+05	5.702E+00
	time	1.969E-01	3.147E-01	4.701E+02	1.063E+00	4.859E-01	1.211E+00	5.525E-01	1.748E-01
	p-val	3.125E-08	1.399E-03	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	6.000E+00	2.000E+00	5.000E+00	3.000E+00	4.000E+00	7.000E+00	8.000E+00	1.000E+00

Table 3. CEC 2015 – 10-dim-2

	Metric	ao	de	pso	HipO	LEA	CMAES	GA	CrPO
Fnc9	worst	1.145E+02	1.005E+02	2.216E+02	1.011E+02	1.004E+02	1.684E+02	1.023E+02	1.003E+02
	mean	1.073E+02	1.003E+02	1.084E+02	1.007E+02	1.003E+02	1.322E+02	1.008E+02	1.002E+02
	best	1.025E+02	1.002E+02	1.003E+02	1.004E+02	1.002E+02	1.087E+02	1.003E+02	1.001E+02
	std	3.040E+00	7.898E-02	2.537E+01	1.814E-01	5.182E-02	1.463E+01	4.749E-01	3.624E-02
	time	2.171E-01	3.568E-01	4.856E+02	1.043E+00	5.349E-01	1.261E+00	5.728E-01	1.848E-01
	p-val	3.125E-08	1.075E-03	3.125E-08	3.125E-08	5.284E-06	3.125E-08	3.125E-08	ref
	rank	6.000E+00	2.000E+00	7.000E+00	4.000E+00	3.000E+00	8.000E+00	5.000E+00	1.000E+00
Fnc10	worst	8.140E+04	3.725E+02	5.868E+04	2.720E+04	7.078E+03	1.402E+06	4.420E+05	2.191E+02
	mean	1.671E+04	2.443E+02	4.586E+03	7.387E+03	1.708E+03	3.861E+05	5.402E+04	2.173E+02
	best	3.015E+03	1.939E+02	3.680E+02	1.125E+03	3.030E+02	1.131E+04	8.252E+02	2.165E+02
	std	1.644E+04	4.158E+01	1.255E+04	7.550E+03	1.556E+03	4.884E+05	1.078E+05	1.118E+00
	time	3.602E-01	4.766E-01	5.954E+02	1.256E+00	6.782E-01	1.416E+00	6.855E-01	3.182E-01
	p-val	3.125E-08	4.125E-05	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	6.000E+00	2.000E+00	4.000E+00	5.000E+00	3.000E+00	8.000E+00	7.000E+00	1.000E+00
Fnc11	worst	3.222E+02	3.007E+02	5.544E+02	3.059E+02	3.004E+02	7.106E+02	5.836E+02	3.001E+02
	mean	2.877E+02	2.718E+02	3.132E+02	3.034E+02	7.704E+01	5.104E+02	3.251E+02	2.298E+02
	best	4.656E+01	1.317E+00	1.105E+01	3.006E+02	4.375E+00	3.077E+02	1.195E+01	3.349E+00
	std	6.923E+01	8.772E+01	9.966E+01	1.660E+00	1.248E+02	1.180E+02	1.524E+02	1.256E+02
	time	9.405E-01	9.619E-01	9.335E+02	1.847E+00	1.299E+00	1.893E+00	1.236E+00	8.479E-01
	p-val	2.232E-06	3.645E-05	2.839E-07	3.086E-08	1.993E-01	3.082E-08	1.501E-05	ref
	rank	4.000E+00	3.000E+00	6.000E+00	5.000E+00	1.000E+00	8.000E+00	7.000E+00	2.000E+00
Fnc12	worst	1.317E+03	1.306E+03	1.328E+03	1.311E+03	1.305E+03	1.352E+03	1.314E+03	1.302E+03

	mean	1.312E+03	1.302E+03	1.307E+03	1.306E+03	1.303E+03	1.325E+03	1.307E+03	1.301E+03
	best	1.307E+03	1.300E+03	1.303E+03	1.303E+03	1.302E+03	1.302E+03	1.302E+03	1.301E+03
	std	2.411E+00	1.210E+00	6.080E+00	2.072E+00	9.505E-01	1.666E+01	3.011E+00	3.648E-01
	time	2.407E-01	3.859E-01	5.179E+02	1.103E+00	5.599E-01	1.326E+00	6.061E-01	2.168E-01
	p-val	3.125E-08	1.188E-01	3.125E-08	3.125E-08	1.273E-07	1.366E-06	8.407E-08	ref
	rank	7.000E+00	2.000E+00	6.000E+00	4.000E+00	3.000E+00	8.000E+00	5.000E+00	1.000E+00
Fnc13	worst	4.632E+01	3.480E+01	5.025E+01	4.791E+01	4.179E+01	4.830E+01	4.246E+01	3.468E+01
	mean	4.356E+01	3.038E+01	4.341E+01	4.085E+01	3.603E+01	4.673E+01	3.834E+01	2.734E+01
	best	4.016E+01	2.741E+01	3.566E+01	3.440E+01	3.020E+01	4.397E+01	3.008E+01	2.352E+01
	std	1.737E+00	2.220E+00	4.195E+00	3.489E+00	3.149E+00	1.038E+00	3.037E+00	2.738E+00
	time	2.692E-01	3.662E-01	5.058E+02	1.127E+00	5.538E-01	1.345E+00	5.928E-01	2.145E-01
	p-val	3.125E-08	1.075E-03	3.125E-08	3.607E-08	9.659E-08	3.125E-08	5.522E-08	ref
	rank	7.000E+00	2.000E+00	6.000E+00	5.000E+00	3.000E+00	8.000E+00	4.000E+00	1.000E+00
Fnc14	worst	7.240E+03	4.251E+03	1.241E+04	7.039E+03	1.003E+04	1.148E+04	1.193E+04	7.037E+03
	mean	5.170E+03	3.021E+03	7.764E+03	5.499E+03	4.907E+03	8.623E+03	5.786E+03	2.590E+03
	best	3.422E+03	2.936E+03	3.097E+03	1.011E+02	1.000E+02	4.664E+03	3.085E+03	1.000E+02
	std	1.372E+03	2.759E+02	2.536E+03	2.031E+03	2.880E+03	1.980E+03	2.120E+03	2.383E+03
	time	3.471E-01	3.663E-01	5.244E+02	1.144E+00	6.008E-01	1.349E+00	5.835E-01	2.382E-01
	p-val	6.658E-05	5.787E-01	1.619E-06	9.441E-04	9.325E-03	1.324E-07	1.415E-05	ref
	rank	4.000E+00	2.000E+00	7.000E+00	5.000E+00	3.000E+00	8.000E+00	6.000E+00	1.000E+00
Fnc15	worst	1.465E+02	1.000E+02	1.293E+02	1.000E+02	1.000E+02	9.211E+02	1.591E+02	1.000E+02
	mean	1.365E+02	1.000E+02	1.201E+02	1.000E+02	1.000E+02	2.373E+02	1.186E+02	1.000E+02
	best	1.284E+02	1.000E+02	1.090E+02	1.000E+02	1.000E+02	1.000E+02	1.073E+02	1.000E+02
	std	4.945E+00	6.345E-03	7.740E+00	3.260E-03	1.111E-03	2.265E+02	1.070E+01	1.160E-14
	time	1.084E+00	9.569E-01	9.695E+02	1.935E+00	1.353E+00	1.906E+00	1.274E+00	8.683E-01
	p-val	1.581E-08	3.169E-03	1.581E-08	1.581E-08	1.581E-08	2.117E-06	1.581E-08	ref
	rank	7.000E+00	3.000E+00	6.000E+00	4.000E+00	2.000E+00	8.000E+00	5.000E+00	1.000E+00

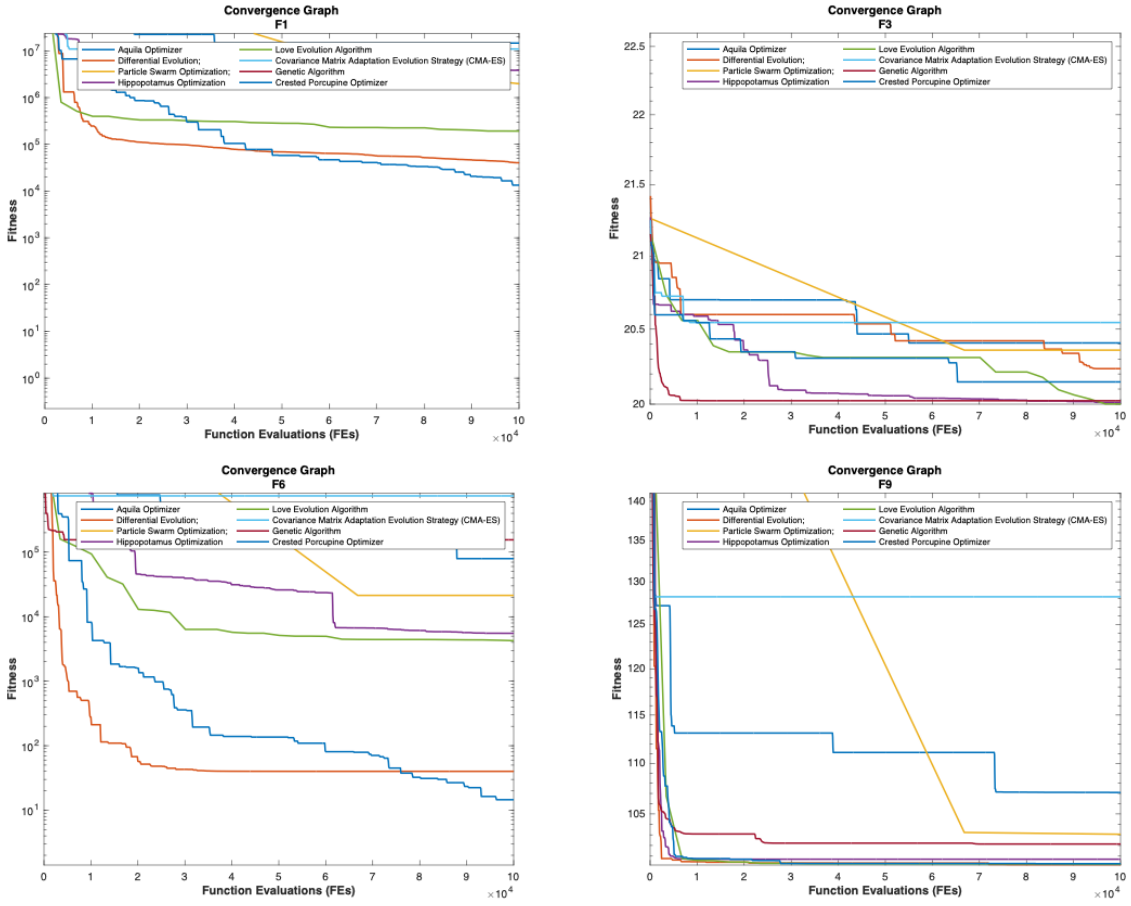


Figure 1. Convergence curves of the F1 (Unimodal), F3 (Simple Multimodal), F6 (Hybrid), and F9 (Composition) functions on the 10-dimensional CEC 2015 benchmark set.

Table 4. CEC 2015 – 30-dim-1

Func.	Metric	ao	de	pso	HipO	LEA	CMAES	GA	CrPO
	worst	9.109E+08	1.798E+06	8.487E+08	1.487E+08	1.833E+06	1.058E+09	9.561E+07	7.892E+04
	mean	5.019E+08	5.899E+05	1.689E+08	5.940E+07	1.037E+06	6.956E+08	5.082E+07	3.226E+04
	best	2.334E+08	1.280E+05	2.504E+07	1.660E+07	4.737E+05	3.371E+08	1.689E+07	3.731E+03
Fnc1	std	1.862E+08	4.116E+05	1.707E+08	3.319E+07	3.906E+05	2.090E+08	2.158E+07	2.233E+04
	time	8.783E-01	1.273E+00	8.843E+02	6.240E+00	2.387E+00	7.338E+00	3.575E+00	6.382E-01
	p-val	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	7.000E+00	2.000E+00	6.000E+00	5.000E+00	3.000E+00	8.000E+00	4.000E+00	1.000E+00
	worst	6.294E+10	2.808E+09	2.653E+10	1.971E+05	1.611E+04	4.397E+10	6.697E+09	4.674E-01
	mean	4.004E+10	1.367E+08	1.640E+10	2.992E+04	8.771E+03	3.270E+10	3.061E+09	2.298E-02
	best	2.578E+10	6.956E+01	9.891E+09	1.484E+03	6.052E+02	4.660E+03	6.129E+08	1.059E-06
Fnc2	std	7.742E+09	5.974E+08	4.402E+09	5.871E+04	4.477E+03	9.994E+09	1.588E+09	9.941E-02
	time	7.313E-01	1.158E+00	9.321E+02	5.777E+00	2.121E+00	7.190E+00	3.491E+00	5.314E-01
	p-val	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	8.000E+00	4.000E+00	6.000E+00	3.000E+00	2.000E+00	7.000E+00	5.000E+00	1.000E+00
	worst	2.110E+01	2.101E+01	2.081E+01	2.005E+01	2.091E+01	2.095E+01	2.018E+01	2.000E+01
	mean	2.103E+01	2.075E+01	2.069E+01	2.001E+01	2.045E+01	2.087E+01	2.010E+01	2.000E+01
	best	2.095E+01	2.040E+01	2.046E+01	2.000E+01	2.001E+01	2.078E+01	2.002E+01	2.000E+01
Fnc3	std	3.724E-02	1.916E-01	9.347E-02	1.498E-02	2.384E-01	5.271E-02	4.650E-02	2.946E-06
	time	7.625E-01	1.290E+00	9.953E+02	5.916E+00	2.231E+00	7.297E+00	3.474E+00	6.568E-01
	p-val	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	8.000E+00	6.000E+00	5.000E+00	2.000E+00	4.000E+00	7.000E+00	3.000E+00	1.000E+00
	worst	3.869E+02	6.229E+01	2.737E+02	2.527E+02	2.169E+02	3.601E+02	2.007E+02	7.562E+01
	mean	3.519E+02	3.899E+01	2.372E+02	2.193E+02	1.441E+02	1.762E+02	1.404E+02	5.373E+01
	best	3.215E+02	1.921E+01	2.087E+02	1.741E+02	6.965E+01	1.492E+01	8.745E+01	3.880E+01
Fnc4	std	1.800E+01	1.146E+01	1.852E+01	2.165E+01	3.759E+01	1.272E+02	2.629E+01	9.782E+00
	time	7.872E-01	1.232E+00	1.035E+03	5.909E+00	2.236E+00	7.388E+00	3.534E+00	6.523E-01
	p-val	3.125E-08	1.611E-04	3.125E-08	3.125E-08	4.159E-08	6.630E-02	3.125E-08	ref
	rank	8.000E+00	1.000E+00	7.000E+00	6.000E+00	4.000E+00	5.000E+00	3.000E+00	2.000E+00
	worst	8.732E+03	6.618E+03	6.020E+03	4.333E+03	4.441E+03	7.668E+03	4.332E+03	3.299E+03
	mean	7.650E+03	3.912E+03	4.699E+03	3.663E+03	3.394E+03	7.205E+03	3.414E+03	2.107E+03
	best	6.646E+03	2.267E+03	3.720E+03	3.045E+03	1.860E+03	6.852E+03	1.875E+03	1.084E+03
Fnc5	std	4.943E+02	1.041E+03	7.006E+02	3.739E+02	6.549E+02	2.400E+02	5.187E+02	5.584E+02
	time	8.675E-01	1.375E+00	9.993E+02	6.120E+00	2.411E+00	7.378E+00	3.546E+00	6.934E-01
	p-val	3.125E-08	3.280E-07	3.125E-08	9.659E-08	4.154E-06	3.109E-08	9.325E-07	ref
	rank	8.000E+00	5.000E+00	6.000E+00	4.000E+00	2.000E+00	7.000E+00	3.000E+00	1.000E+00
	worst	7.373E+07	2.425E+05	6.711E+07	7.521E+06	2.496E+05	9.770E+07	5.035E+07	2.442E+04
	mean	1.854E+07	6.619E+04	1.079E+07	9.398E+05	1.169E+05	4.586E+07	1.558E+07	6.932E+03
	best	3.553E+06	5.663E+03	3.774E+05	7.700E+04	1.693E+04	1.374E+07	3.113E+06	6.763E+02
Fnc6	std	1.351E+07	5.816E+04	1.657E+07	1.574E+06	6.514E+04	1.977E+07	1.222E+07	6.339E+03
	time	8.013E-01	1.248E+00	9.440E+02	5.945E+00	2.257E+00	7.322E+00	3.549E+00	6.521E-01
	p-val	3.125E-08	4.876E-07	3.125E-08	3.125E-08	4.794E-08	3.125E-08	3.125E-08	ref
	rank	7.000E+00	2.000E+00	5.000E+00	4.000E+00	3.000E+00	8.000E+00	6.000E+00	1.000E+00
	worst	1.488E+02	1.178E+01	1.460E+02	9.562E+01	8.083E+01	3.324E+02	1.315E+02	4.560E+00
	mean	1.081E+02	8.524E+00	4.868E+01	5.325E+01	2.166E+01	2.051E+02	4.330E+01	3.179E+00
	best	5.803E+01	5.879E+00	3.136E+01	2.347E+01	7.614E+00	1.077E+02	1.493E+01	1.801E+00
Fnc7	std	2.098E+01	1.578E+00	2.626E+01	2.776E+01	1.914E+01	6.325E+01	3.407E+01	6.525E-01
	time	2.179E+00	2.383E+00	1.259E+03	7.290E+00	3.600E+00	8.315E+00	4.582E+00	1.761E+00
	p-val	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	7.000E+00	2.000E+00	5.000E+00	6.000E+00	3.000E+00	8.000E+00	4.000E+00	1.000E+00
	worst	7.652E+06	1.451E+05	1.487E+07	9.036E+05	9.439E+04	3.465E+07	2.040E+07	5.900E+03
	mean	2.991E+06	5.349E+04	1.510E+06	1.740E+05	5.954E+04	1.733E+07	5.609E+06	1.874E+03
	best	9.406E+05	3.325E+03	1.244E+05	3.202E+04	2.086E+04	4.733E+06	8.027E+05	7.179E+01
Fnc8	std	1.736E+06	4.643E+04	3.092E+06	1.966E+05	2.369E+04	7.745E+06	5.256E+06	1.522E+03
	time	7.964E-01	1.290E+00	1.020E+03	5.872E+00	2.231E+00	7.307E+00	3.581E+00	6.308E-01
	p-val	3.125E-08	9.659E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	3.125E-08	ref
	rank	6.000E+00	2.000E+00	5.000E+00	4.000E+00	3.000E+00	8.000E+00	7.000E+00	1.000E+00

Table 5. CEC 2015 – 30-dim-2

Func.	Metric	ao	de	pso	HipO	LEA	CMAES	GA	CrPO
	worst	5.930E+02	1.055E+02	4.875E+02	4.545E+02	1.036E+02	5.813E+02	2.992E+02	1.034E+02
	mean	2.625E+02	1.032E+02	2.627E+02	1.257E+02	1.032E+02	3.307E+02	1.286E+02	1.029E+02
	best	2.020E+02	1.023E+02	1.460E+02	1.069E+02	1.029E+02	1.951E+02	1.069E+02	1.024E+02
Fnc9	std	7.810E+01	7.649E-01	1.115E+02	7.355E+01	1.696E-01	1.412E+02	3.927E+01	2.162E-01
	time	9.797E-01	1.407E+00	1.041E+03	6.074E+00	2.485E+00	7.411E+00	3.668E+00	8.479E-01
	p-val	3.125E-08	2.576E-01	3.125E-08	3.125E-08	5.700E-05	3.125E-08	3.125E-08	ref
	rank	6.000E+00	3.000E+00	7.000E+00	4.000E+00	2.000E+00	8.000E+00	5.000E+00	1.000E+00
	worst	1.665E+07	4.382E+05	2.541E+07	5.509E+06	7.897E+04	7.973E+07	2.682E+07	1.203E+04
	mean	1.021E+07	1.098E+05	5.540E+06	1.595E+06	4.313E+04	4.292E+07	9.286E+06	4.412E+03
	best	4.285E+06	5.665E+03	9.549E+04	4.607E+04	6.436E+03	1.197E+07	5.222E+05	7.488E+02
Fnc10	std	3.048E+06	1.072E+05	7.200E+06	1.345E+06	1.782E+04	1.608E+07	7.420E+06	3.267E+03
	time	2.381E+00	2.854E+00	1.344E+03	7.607E+00	3.950E+00	8.650E+00	4.801E+00	2.121E+00
	p-val	3.125E-08	1.917E-07	3.125E-08	3.125E-08	6.356E-08	3.125E-08	3.125E-08	ref
	rank	7.000E+00	3.000E+00	5.000E+00	4.000E+00	2.000E+00	8.000E+00	6.000E+00	1.000E+00
	worst	1.526E+03	7.636E+02	1.583E+03	1.546E+03	1.131E+03	1.551E+03	1.163E+03	8.066E+02
	mean	1.310E+03	6.228E+02	1.255E+03	8.066E+02	7.756E+02	1.478E+03	1.034E+03	4.366E+02
	best	7.331E+02	3.040E+02	4.101E+02	3.114E+02	3.042E+02	1.358E+03	9.481E+02	3.009E+02
Fnc11	std	2.812E+02	1.112E+02	2.254E+02	5.073E+02	3.068E+02	4.569E+01	5.454E+01	1.546E+02
	time	7.979E+00	6.957E+00	2.345E+03	1.320E+01	9.572E+00	1.331E+01	9.701E+00	6.547E+00
	p-val	3.607E-08	4.597E-05	1.273E-07	1.188E-02	6.341E-05	3.125E-08	3.125E-08	ref
	rank	7.000E+00	2.000E+00	6.000E+00	4.000E+00	3.000E+00	8.000E+00	5.000E+00	1.000E+00
	worst	1.377E+03	1.312E+03	1.390E+03	1.319E+03	1.313E+03	1.372E+03	1.317E+03	1.308E+03
	mean	1.357E+03	1.307E+03	1.365E+03	1.313E+03	1.310E+03	1.358E+03	1.312E+03	1.306E+03
	best	1.345E+03	1.305E+03	1.336E+03	1.309E+03	1.307E+03	1.350E+03	1.308E+03	1.305E+03
Fnc12	std	6.908E+00	1.659E+00	1.234E+01	2.179E+00	1.523E+00	5.630E+00	2.061E+00	8.260E-01
	time	1.363E+00	1.885E+00	1.142E+03	6.402E+00	2.864E+00	7.740E+00	3.937E+00	1.144E+00
	p-val	3.125E-08	4.689E-02	3.125E-08	3.125E-08	7.312E-08	3.125E-08	3.125E-08	ref
	rank	6.000E+00	2.000E+00	8.000E+00	5.000E+00	3.000E+00	7.000E+00	4.000E+00	1.000E+00
	worst	1.464E+02	1.198E+02	1.453E+02	1.468E+02	1.300E+02	1.407E+02	1.254E+02	1.129E+02
	mean	1.425E+02	1.104E+02	1.358E+02	1.365E+02	1.220E+02	1.383E+02	1.174E+02	9.800E+01
	best	1.388E+02	9.880E+01	1.223E+02	1.246E+02	1.109E+02	1.361E+02	1.067E+02	8.787E+01
Fnc13	std	2.155E+00	5.118E+00	7.279E+00	5.695E+00	4.542E+00	8.553E-01	5.068E+00	6.483E+00
	time	1.436E+00	1.862E+00	1.063E+03	6.213E+00	2.876E+00	7.511E+00	3.814E+00	1.149E+00
	p-val	3.125E-08	5.284E-06	3.125E-08	3.125E-08	3.607E-08	3.098E-08	6.356E-08	ref
	rank	8.000E+00	2.000E+00	5.000E+00	6.000E+00	4.000E+00	7.000E+00	3.000E+00	1.000E+00
	worst	7.633E+04	3.464E+04	1.012E+05	4.794E+04	3.717E+04	5.059E+04	3.917E+04	3.796E+04
	mean	5.742E+04	3.171E+04	5.960E+04	3.905E+04	3.438E+04	4.633E+04	3.630E+04	3.372E+04
	best	4.647E+04	3.115E+04	4.532E+04	3.218E+04	3.139E+04	4.580E+04	3.369E+04	3.148E+04
Fnc14	std	7.805E+03	7.440E+02	1.271E+04	3.861E+03	1.335E+03	1.017E+03	1.275E+03	1.445E+03
	time	1.589E+00	1.885E+00	1.145E+03	6.612E+00	3.145E+00	7.804E+00	4.136E+00	1.258E+00
	p-val	3.125E-08	2.883E-06	3.125E-08	4.687E-06	7.011E-02	1.023E-08	6.704E-06	ref
	rank	7.000E+00	1.000E+00	8.000E+00	5.000E+00	3.000E+00	6.000E+00	4.000E+00	2.000E+00
	worst	1.620E+04	1.052E+02	1.217E+03	1.677E+02	1.000E+02	1.381E+04	8.818E+02	1.000E+02
	mean	5.176E+03	1.005E+02	3.699E+02	1.355E+02	1.000E+02	2.251E+03	2.306E+02	1.000E+02
	best	1.014E+03	1.000E+02	1.391E+02	1.228E+02	1.000E+02	1.000E+02	1.196E+02	1.000E+02
Fnc15	std	3.400E+03	1.136E+00	3.162E+02	1.353E+01	6.863E-03	3.024E+03	1.773E+02	1.421E-14
	time	8.169E+00	7.390E+00	2.354E+03	1.347E+01	9.934E+00	1.336E+01	9.801E+00	6.843E+00
	p-val	3.306E-09	3.306E-09	3.306E-09	3.306E-09	3.306E-09	1.237E-08	3.306E-09	ref
	rank	8.000E+00	3.000E+00	6.000E+00	4.000E+00	2.000E+00	7.000E+00	5.000E+00	1.000E+00

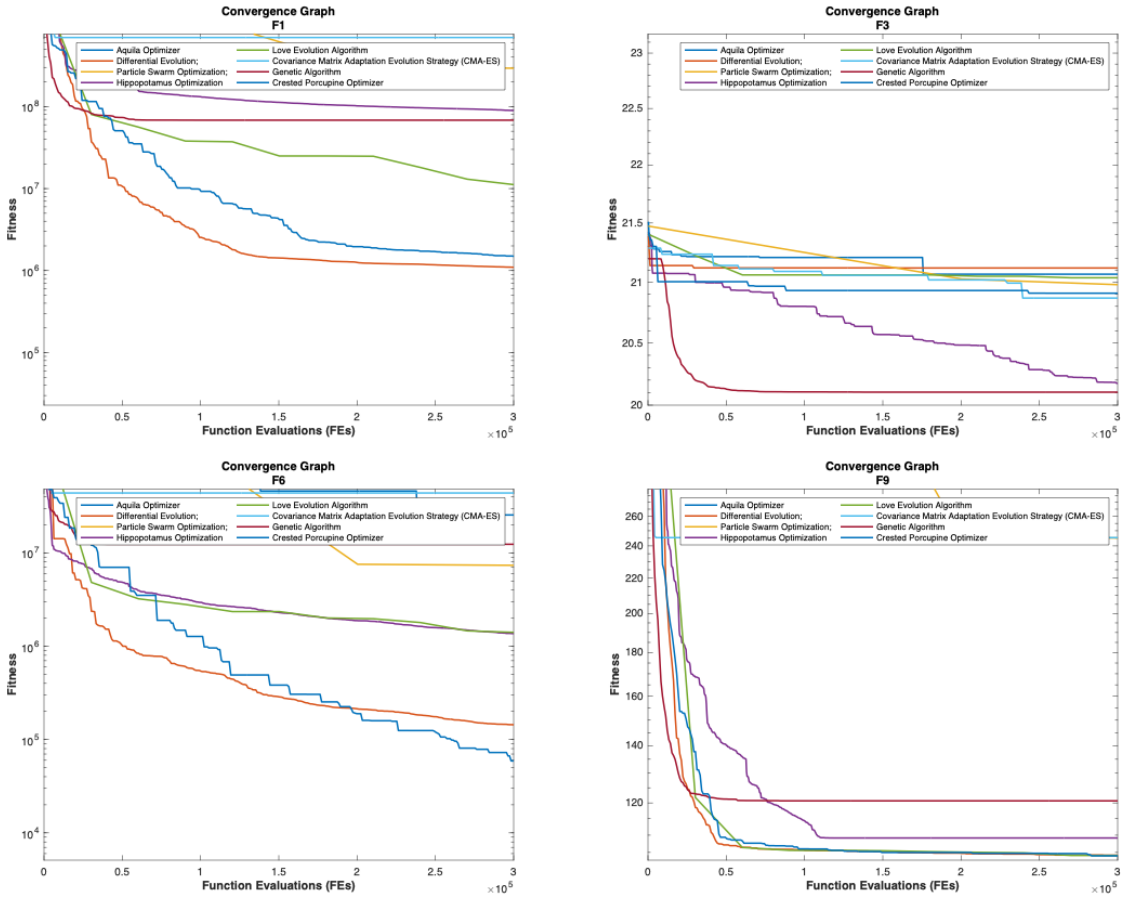


Figure 2 Convergence curves of the F1 (Unimodal), F3 (Simple Multimodal), F6 (Hybrid), and F9 (Composition) functions on the 30-dimensional CEC 2015 benchmark set

4. CONCLUSION

This study presented an in-depth performance evaluation of the (CrPO), a recently developed metaheuristic inspired by the hierarchical defensive behaviors of crested porcupines. The algorithm was tested on the CEC2015 benchmark functions, covering a wide range of problem types including unimodal, multimodal, hybrid, and composition functions. CrPO utilizes four biologically inspired defense mechanisms—sight, sound, odor, and physical attack—modeled as exploration and exploitation strategies within the search space. Furthermore, a cyclic population reduction mechanism was incorporated to enhance convergence speed and preserve population diversity.

The experimental results, based on 51 independent runs for each function in 10 and 30 dimensions, demonstrated that CrPO consistently outperformed several well-established algorithms, including Differential Evolution (DE), Love Evolution Algorithm (LEA), Hippopotamus Optimizer (HipO), Aquila Optimizer (AO) Covariance Matrix Adaptation Evolution Strategy (CMA-ES), Particle Swarm Optimization (PSO) and, Genetic Algorithm. CrPO achieved the best mean ranks across most benchmark functions. The superiority of CrPO was statistically validated using Friedman's mean rank test.

In conclusion, the CrPO algorithm proves to be a robust and competitive optimization tool capable of effectively solving complex numerical optimization problems. Future work will focus on extending the algorithm to constrained, dynamic, and multi-objective optimization problems, and exploring potential hybridizations with other optimization strategies.

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